

The Political Economy of Proximity to the Standard Nuclear oliGARCHy

Demonstrated Power, Strategic Distance, and a Quasi-Natural Experiment

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops an interdisciplinary framework for studying the political economy of geographic and strategic proximity to the Standard Nuclear oliGARCHy (SNoG henceforth). The central identification idea is that geography may be predetermined while the strategic meaning of geography changes discontinuously after a salient demonstration of intervention capability. The historical motivation is the United States' atomic bombings of Hiroshima on August 6, 1945, and Nagasaki on August 9, 1945. These events did not merely destroy two cities and contribute to the termination of the Pacific War; they publicly demonstrated a military technology whose existence and operational consequences could induce governments to revise beliefs about security, deterrence, vulnerability, alliance formation, military expenditure, investment, fiscal allocation, and technological competition.

We formalize this mechanism as the interaction of predetermined strategic proximity with an intervention-induced information shock. The resulting empirical architecture combines continuous-treatment difference-in-differences, event studies, spatial econometrics, randomization inference, synthetic controls, heterogeneous treatment effects, Bayesian belief updating, and machine-learning-assisted causal analysis. A political-economic model shows that proximity to demonstrated power need not have a single sign: proximity to an adversarial power may increase defensive effort, whereas proximity to a credible protector may reduce indigenous defense requirements through a security externality.

The framework is extended using concepts from economics, finance, physics, chemistry, biology, psychology, philosophy, political science, international relations, warfare economics, statistics, computer science, artificial intelligence, network science, information theory, and control theory. Importantly, the paper distinguishes a proposed empirical design from completed empirical evidence: causal claims require data, identification diagnostics, robustness tests, and historical validation. The resulting framework interprets SNoG proximity as a problem of demonstrated power, belief updating, strategic distance, network transmission, and endogenous political-economic adaptation.

The paper ends with "The End"

Keywords: Standard Nuclear oliGARCHy, nuclear weapons, Hiroshima, Nagasaki, political economy, deterrence, geopolitics, difference-in-differences, quasi-natural experiment, Bayesian updating, spatial economics, warfare economics, causal inference.

Contents

1	Introduction	4
2	Historical Motivation: Hiroshima, Nagasaki, and Demonstrated Power	5
3	The Concept of SNoG Proximity	5
4	A Theory of Demonstrated Power	6
5	The SNoG Proximity Hypothesis	6
6	A Basic Proposition	7

7	Protection and the Possibility of Defense Crowding Out	8
8	The Quasi-Natural Experiment	8
9	Potential Outcomes	8
10	Identification Assumptions	9
11	Event-Study Design	9
12	Strategic Distance Decay	10
13	Alignment Heterogeneity	10
14	Political Science and International Relations	10
15	Game-Theoretic Deterrence	11
16	Economics of Fiscal Reallocation	11
17	Warfare Economics	12
18	Finance and Asset Pricing	12
19	Welfare Economics	13
20	Psychology: Salience, Fear, and Belief Updating	13
21	Psychiatry and Collective Stress	13
22	Philosophy and the Ethics of Demonstrated Power	14
23	Physics: Distance, Reach, and Propagation	14
24	Chemistry: Thresholds and Strategic Activation	14
25	Biology and Evolutionary Adaptation	15
26	Network Science	15
27	Information Theory	15
28	Control Theory	16
29	Data Architecture	16
30	Outcome Families	16
31	Spatial Econometrics	17
32	Randomization Inference with Pseudo-SNoG Locations	17
33	Placebo Dates	18
34	Synthetic Control Methods	18
35	Machine Learning and Heterogeneous Treatment Effects	18
36	Double Machine Learning	19
37	A Bayesian SNoG Model	19
38	A General Equilibrium Extension	20

39 Arms-Race Dynamics	21
40 Technology and Innovation	21
41 Supply Chains and Economic Geography	22
42 Migration and Demography	22
43 An Integrated Structural System	22
44 Directed Acyclic Graph	23
45 Vector Graphic: Distance-Decay Function	23
46 Vector Graphic: Threat versus Protection	24
47 Empirical Hypotheses	24
48 Historical Validation Strategy	24
49 Threats to Identification	25
50 Robustness Architecture	26
51 Pseudo-Code for the Empirical Pipeline	26
52 Pseudo-Code for AI-Assisted Heterogeneity Analysis	27
53 Simulation Framework	28
54 Falsifiability	28
55 Interpretation of Null Results	29
56 External Validity	29
57 Limits of the Hiroshima–Nagasaki Analogy	30
58 Ethical and Legal Boundaries	30
59 A Unified SNoG Proximity Index	30
60 A Structural SNoG Response Function	31
61 Main Theoretical Result	31
62 Interpretive Matrix	32
63 Research Design Decision Tree	33
64 Discussion	33
65 Conclusion	34

List of Figures

1	Conceptual causal architecture of demonstrated power and strategic proximity.	23
2	Illustrative exponential distance decay, $\beta(d) = \beta_0 e^{-\rho d}$, normalized to $\beta_0 = 1$	23
3	Illustrative transition from threat-dominated to protection-dominated proximity effects. .	24
4	Research-design decision tree for a SNoG proximity study.	33

List of Tables

1	Proposed Empirical Variables	16
2	Principal Empirical Hypotheses	24
3	Identification Threats and Responses	25
4	Possible Political-Economic Regimes	32

1 Introduction

Political geography is persistent, but the strategic meaning of geography is not.

A country may remain at exactly the same latitude and longitude while the economic and political consequences of its location change abruptly. New military technologies, alliances, doctrines, sanctions regimes, trade institutions, military bases, nuclear umbrellas, or demonstrations of coercive capacity can transform distance from a geographic fact into an economically relevant treatment.

This paper studies that proposition through the concept of proximity to the Standard Nuclear oligARCHy (SNoG).

The central causal structure is

$$\boxed{\text{Predetermined Geography} \times \text{Demonstrated Capability} \longrightarrow \text{Political-Economic Adjustment.}}$$

The motivating historical analogy is the United States' use of atomic weapons against Hiroshima and Nagasaki in August 1945. The historical episode is not treated as equivalent to a hypothetical SNoG intervention. Rather, it illustrates an important identification principle: an observable intervention can reveal information about previously uncertain military capabilities and willingness to employ them.

Let T_0 denote the date of such a strategic revelation. Let d_i denote the distance of political unit i from the relevant center, frontier, military installation, alliance system, or strategic network.

Geography may satisfy

$$d_i(T_0^-) = d_i(T_0^+),$$

while perceived strategic exposure satisfies

$$R_i(T_0^-) \neq R_i(T_0^+).$$

The empirical treatment therefore does not consist of physical movement. Instead, it consists of a discontinuous change in the strategic importance of pre-existing spatial relationships.

The paper makes five principal contributions.

First, it develops a theory of *demonstrated power*. Capabilities affect political behavior not only through their physical existence but also through beliefs about their availability, credibility, effectiveness, and probability of use.

Second, it develops a continuous spatial treatment:

$$P_i = \sum_{j=1}^J \frac{w_j}{(d_{ij} + d_0)^\rho},$$

where P_i is strategic proximity, w_j is the strategic importance of SNoG node j , d_{ij} is distance, $d_0 > 0$ prevents singularities, and $\rho > 0$ determines distance decay.

Third, it allows proximity to generate either threat or protection.

Fourth, it develops a quasi-experimental empirical architecture based on

$$P_i \times \text{Post}_t.$$

Fifth, it integrates the political-economic framework with insights from multiple scientific disciplines without confusing analogy with causal proof.

2 Historical Motivation: Hiroshima, Nagasaki, and Demonstrated Power

On August 6, 1945, the United States dropped an atomic bomb on Hiroshima. A second atomic bomb was dropped on Nagasaki on August 9, 1945. Japan announced its acceptance of the Allied surrender terms on August 15, and the formal surrender occurred on September 2, 1945.

The events transformed international politics because nuclear weapons became not merely theoretical technologies but demonstrated instruments of warfare.

The distinction can be represented as

$$\text{latent capability} \neq \text{publicly demonstrated capability}.$$

Suppose a foreign government believes that a state possesses some latent intervention parameter θ . Before demonstration,

$$\theta \sim \pi_0(\theta).$$

Following an observed intervention E ,

$$\pi_1(\theta) = \frac{\Pr(E \mid \theta)\pi_0(\theta)}{\int \Pr(E \mid \tilde{\theta})\pi_0(\tilde{\theta})d\tilde{\theta}}.$$

If the event is more likely under high values of θ , then the posterior mean may satisfy

$$\mathbb{E}_{\pi_1}[\theta] > \mathbb{E}_{\pi_0}[\theta].$$

This does not imply that every government updates identically. Political leaders possess different information sets, preferences, intelligence capabilities, doctrines, and alliance relationships.

The general mechanism is therefore

$$E \rightarrow \text{information} \rightarrow \text{belief updating} \rightarrow \text{policy response}.$$

3 The Concept of SNoG Proximity

Definition 1 (Geographic Proximity). *For country i and SNoG node j , geographic proximity is a decreasing function of distance:*

$$p_{ij} = g(d_{ij}), \quad g'(d) < 0.$$

A convenient functional form is

$$g(d) = \frac{1}{(d + d_0)^\rho},$$

with $d_0 > 0$ and $\rho > 0$.

Definition 2 (Strategic Proximity). *Strategic proximity is*

$$P_i = \sum_{j=1}^J w_j g(d_{ij}),$$

where $w_j \geq 0$ represents the strategic relevance of node j .

Strategic proximity differs from Euclidean distance because transportation, terrain, missile range, maritime access, air corridors, alliances, communications networks, and economic dependence may alter effective distance.

We therefore define a generalized strategic metric

$$D_{ij}^S = \omega_1 D_{ij}^{geo} + \omega_2 D_{ij}^{log} + \omega_3 D_{ij}^{econ} + \omega_4 D_{ij}^{dip} + \omega_5 D_{ij}^{net},$$

where the terms represent geographic, logistical, economic, diplomatic, and network distance.

The corresponding proximity measure is

$$P_{ij}^S = \frac{1}{(D_{ij}^S + d_0)^\rho}.$$

4 A Theory of Demonstrated Power

Let C_i denote civilian consumption, I_i productive public investment, D_i defense expenditure, and A_i strategic alignment.

Government i solves

$$\max_{C_i, I_i, D_i, A_i} U_i(C_i, I_i, D_i, A_i; R_i)$$

subject to

$$C_i + I_i + D_i + \Psi(A_i) = R_i^f + B_i,$$

where R_i^f denotes fiscal resources and B_i borrowing.

Let perceived external strategic pressure be

$$R_i^S = P_i \mathbb{E}[\theta \mid \mathcal{I}_i] \Xi_i,$$

where Ξ_i describes whether the relationship with SNoG is adversarial, neutral, or protective.

A simple representation is

$$\Xi_i = 1 - 2A_i,$$

with $A_i \in [0, 1]$. Then

$$\Xi_i > 0$$

indicates net threat and

$$\Xi_i < 0$$

indicates net protection.

The marginal return to defense may satisfy

$$\frac{\partial^2 U_i}{\partial D_i \partial R_i^S} > 0.$$

Hence greater strategic threat increases the marginal value of defensive capacity.

5 The SNoG Proximity Hypothesis

Hypothesis 1 (SNoG Proximity Hypothesis). *Following a salient demonstration of SNoG intervention capacity, the magnitude of political-economic adjustment is increasing in strategic proximity to SNoG, conditional on the sign and credibility of the state's security relationship with SNoG.*

This hypothesis is deliberately directional only after conditioning on alignment.

For adversarial states,

$$P_i \uparrow \implies R_i^S \uparrow.$$

For protected states,

$$P_i \uparrow \implies \text{security externality} \uparrow.$$

The two mechanisms can produce opposite fiscal responses.

6 A Basic Proposition

Consider

$$U(C, D) = u(C) + v(D, R),$$

subject to

$$C + D = Y.$$

Substitution gives

$$\max_D u(Y - D) + v(D, R).$$

The first-order condition is

$$-u'(Y - D) + v_D(D, R) = 0.$$

Proposition 1 (Threat-Induced Defense Response). *Suppose*

$$u'' < 0, \quad v_{DD} < 0, \quad v_{DR} > 0.$$

Then the optimal defense expenditure $D^(R)$ satisfies*

$$\frac{dD^*}{dR} > 0.$$

Proof. Define

$$F(D, R) = -u'(Y - D) + v_D(D, R).$$

At the optimum,

$$F(D^*, R) = 0.$$

By the implicit function theorem,

$$\frac{dD^*}{dR} = -\frac{F_R}{F_D}.$$

Now

$$F_R = v_{DR} > 0.$$

Furthermore,

$$F_D = u''(Y - D) + v_{DD}(D, R) < 0.$$

Therefore,

$$\frac{dD^*}{dR} = -\frac{v_{DR}}{u'' + v_{DD}} > 0.$$

□

If

$$R = P\mathbb{E}[\theta \mid E],$$

then, under positive marginal effects,

$$\frac{\partial D^*}{\partial P} > 0$$

for a threatened state.

7 Protection and the Possibility of Defense Crowding Out

Suppose domestic security is produced by

$$S_i = f(D_i, G_i),$$

where G_i is the external security guarantee supplied by SNoG.
Let

$$G_i = g(P_i, A_i, C_i),$$

where C_i is guarantee credibility.

If domestic and external defense are substitutes,

$$f_{DG} < 0$$

in the relevant marginal sense.

A sufficiently credible external security guarantee may therefore produce

$$\frac{\partial D_i^*}{\partial P_i} < 0.$$

Thus the theory predicts heterogeneous signs.

This yields the distinction

Proximity to Power $\neq$ Proximity to Threat.

Likewise,

Proximity to Power $\neq$ Proximity to Protection.

The political relationship determines which interpretation dominates.

8 The Quasi-Natural Experiment

Let

$$Post_t = \mathbf{1}\{t \geq T_0\}.$$

The baseline continuous-treatment difference-in-differences model is

$$Y_{it} = \alpha_i + \lambda_t + \beta(P_i \times Post_t) + X'_{it}\gamma + \varepsilon_{it}.$$

Country fixed effects α_i absorb persistent country characteristics.

Time fixed effects λ_t absorb common global shocks.

The coefficient β measures whether units with greater predetermined proximity experience systematically different post-event changes.

The identifying variation is

Cross-sectional proximity $\times$ temporal intervention.

9 Potential Outcomes

Let

$$Y_{it}(z)$$

denote the potential outcome under exposure z .

For continuous treatment,

$$z_{it} = P_i Post_t.$$

A marginal causal response is

$$\tau(p) = \frac{\partial}{\partial p} \mathbb{E}[Y_{it}(p) - Y_{it}(0)].$$

The estimand is not merely a treated-versus-control average treatment effect. Instead, the framework estimates a spatial treatment gradient.

Under a linear approximation,

$$\tau(p) = \beta.$$

Under nonlinear treatment effects,

$$Y_{it} = \alpha_i + \lambda_t + h(P_i)Post_t + X'_{it}\gamma + \varepsilon_{it}.$$

10 Identification Assumptions

Assumption 1 (Predetermined Geography). *Conditional on controls and fixed effects, the relevant geographic variables are not themselves caused by the intervention at T_0 .*

Assumption 2 (Conditional Parallel Trends). *Absent the intervention,*

$$\mathbb{E}[Y_{it}(0) - Y_{i,t-1}(0) \mid P_i, X_i]$$

does not vary systematically with P_i after conditioning on the empirical design.

Assumption 3 (No Differential Anticipation). *Units with different values of P_i do not differentially change the outcome before T_0 because they perfectly anticipate the intervention.*

Assumption 4 (Controlled Interference). *Spatial spillovers are either sufficiently limited or explicitly modeled.*

The final assumption is particularly important because geopolitics is intrinsically networked.

11 Event-Study Design

The dynamic specification is

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k P_i \mathbf{1}\{t - T_0 = k\} + X'_{it}\gamma + \varepsilon_{it}.$$

The omitted period $k = -1$ is the reference period.

Pre-treatment identification requires approximately

$$\beta_k \approx 0 \quad \text{for } k < 0.$$

Post-treatment coefficients trace adjustment:

$$\{\beta_0, \beta_1, \dots, \beta_K\}.$$

The response need not be instantaneous. A useful dynamic model is

$$\beta_k = \beta_\infty (1 - e^{-\kappa k}), \quad k \geq 0,$$

where κ controls adaptation speed.

12 Strategic Distance Decay

A natural hypothesis is that treatment effects decay with distance:

$$\beta(d) = \beta_0 e^{-\rho d}.$$

Then

$$\frac{\partial \beta(d)}{\partial d} = -\rho \beta_0 e^{-\rho d}.$$

For $\beta_0 > 0$ and $\rho > 0$,

$$\frac{\partial \beta(d)}{\partial d} < 0.$$

The half-life distance of the geopolitical effect satisfies

$$\frac{\beta(d_{1/2})}{\beta_0} = \frac{1}{2}.$$

Hence

$$d_{1/2} = \frac{\ln 2}{\rho}.$$

This quantity provides an interpretable measure of the geographic reach of demonstrated power.

13 Alignment Heterogeneity

Let A_i measure pre-treatment alignment with SNoG.

Estimate

$$Y_{it} = \alpha_i + \lambda_t + \beta_1 P_i Post_t + \beta_2 P_i Post_t A_i + X'_{it} \gamma + \varepsilon_{it}.$$

The marginal proximity effect is

$$\frac{\partial Y_{it}}{\partial (P_i Post_t)} = \beta_1 + \beta_2 A_i.$$

A sign reversal occurs at

$$A_i^* = -\frac{\beta_1}{\beta_2},$$

provided $\beta_2 \neq 0$.

This threshold distinguishes the threat-dominated and protection-dominated regimes.

14 Political Science and International Relations

The framework intersects with realism, deterrence theory, balance-of-power theory, alliance theory, security dilemmas, and institutional accounts of international cooperation.

Let national power be

$$\Pi_i = f(M_i, E_i, T_i, N_i, L_i),$$

where

- M_i is military capability,
- E_i is economic capability,
- T_i is technological capability,
- N_i is alliance-network centrality,

- L_i is logistical capability.

A state's strategic response can be represented as

$$a_i^* = \arg \max_{a_i} \mathbb{E}[U_i(a_i, a_{-i}, \Pi) \mid \mathcal{I}_i].$$

The SNoG intervention changes $\mathcal{I}_i$.

Thus,

$$a_i^-(\mathcal{I}_i^-) \neq a_i^+(\mathcal{I}_i^+)$$

even if the state's material endowment remains unchanged.

15 Game-Theoretic Deterrence

Consider two actors: SNoG, denoted S , and state i .

State i chooses either Challenge (C) or Accommodate (A).

SNoG chooses Respond (R) or Not Respond (N).

Before demonstrated intervention, state i believes

$$q_0 = \Pr(R \mid C).$$

After demonstration,

$$q_1 = \Pr(R \mid C, E).$$

If

$$q_1 > q_0,$$

the expected payoff from challenge falls.

Let the payoff from successful challenge be B and the cost of a response be K .

Then

$$\mathbb{E}[U_i(C)] = (1 - q)B - qK.$$

Accommodation yields normalized payoff zero.

Challenge is optimal if

$$(1 - q)B - qK > 0.$$

Equivalently,

$$q < \frac{B}{B + K}.$$

Thus a demonstration can change equilibrium behavior without changing the physical balance of resources if it changes q sufficiently.

16 Economics of Fiscal Reallocation

The government budget constraint is

$$G_i = C_i + I_i + D_i + H_i + S_i,$$

where H_i is health expenditure and S_i social expenditure.

Holding total fiscal capacity fixed,

$$dC_i + dI_i + dD_i + dH_i + dS_i = 0.$$

Therefore,

$$dD_i > 0$$

requires some combination of

$$dC_i < 0, \quad dI_i < 0, \quad dH_i < 0, \quad dS_i < 0,$$

unless taxes, borrowing, seigniorage, or external transfers rise.

The political economy of SNoG proximity therefore extends beyond military budgets.

17 Warfare Economics

Let total mobilizable resources be

$$M_i = Y_i - C_i^{min} - K_i^{maint},$$

where C_i^{min} is minimum politically sustainable civilian consumption and K_i^{maint} is capital maintenance.

Maximum sustainable military effort satisfies

$$D_i \leq M_i.$$

Define the mobilization ratio

$$m_i = \frac{D_i}{Y_i}.$$

A proximity shock may produce

$$\frac{\partial m_i}{\partial P_i} > 0$$

for threatened states.

However, prolonged mobilization may reduce future output:

$$Y_{i,t+1} = F(K_{it}, L_{it}, A_{it}) - \delta_D D_{it},$$

where δ_D captures the opportunity cost or destructive burden of militarization.

Thus the security-production problem is intertemporal.

18 Finance and Asset Pricing

Geopolitical risk can affect asset prices before it affects quantities.

Let sovereign yield be

$$y_{it} = r_t^* + \pi_{it}^{default} + \pi_{it}^{inflation} + \pi_{it}^{liquidity} + \pi_{it}^{geo}.$$

Suppose

$$\pi_{it}^{geo} = \phi R_{it}^S.$$

Then

$$\frac{\partial y_{it}}{\partial P_i} = \phi \frac{\partial R_{it}^S}{\partial P_i}.$$

For equities,

$$V_{it} = \mathbb{E}_t \left[\sum_{s=1}^{\infty} \frac{CF_{i,t+s}}{\prod_{\ell=1}^s (1 + r_{i,t+\ell})} \right].$$

A geopolitical shock can reduce valuation through either expected cash flows or discount rates:

$$dV = \frac{\partial V}{\partial CF} dCF + \frac{\partial V}{\partial r} dr.$$

Empirical financial outcomes can therefore include sovereign spreads, exchange-rate volatility, equity returns, capital flows, credit ratings, and foreign direct investment.

19 Welfare Economics

Let social welfare be

$$W_i = u(C_i) + v(I_i) + h(Sec_i) - \chi(D_i) - \Omega_i.$$

Here Ω_i represents expected destruction, coercion, displacement, political repression, or other social costs associated with insecurity.

The welfare effect of increased SNoG proximity is

$$\frac{dW_i}{dP_i} = u_C \frac{dC_i}{dP_i} + v_I \frac{dI_i}{dP_i} + h_{Sec} \frac{dSec_i}{dP_i} - \chi_D \frac{dD_i}{dP_i} - \frac{d\Omega_i}{dP_i}.$$

No general sign follows.

This is important. A change in measured military security need not imply a Pareto improvement, and a transfer of geopolitical risk from one country to another is not an aggregate welfare gain.

Global welfare is

$$W^G = \sum_i \omega_i W_i.$$

Distributional weights ω_i must therefore be stated explicitly.

20 Psychology: Salience, Fear, and Belief Updating

Strategic behavior is not generated by Bayesian reasoning alone.

Let perceived risk be

$$\hat{R}_i = R_i + \eta_i S_t,$$

where S_t is event salience and η_i is psychological sensitivity.

If agents overweight salient low-probability events, then

$$\hat{p} = w(p),$$

where $w(p)$ is a probability-weighting function.

A government may therefore react according to

$$D_i^* = D^*(w(p_i), K_i).$$

The empirical implication is that highly visible demonstrations can generate responses larger than those predicted by objective probabilities alone.

21 Psychiatry and Collective Stress

Psychiatric concepts should not be mechanically applied to states. However, population-level mental-health outcomes may constitute legitimate consequences of war, perceived insecurity, displacement, and catastrophic events.

Let population psychological burden be

$$H_{it}^{psych} = f(Exposure_{it}, Displacement_{it}, Loss_{it}, Uncertainty_{it}, SocialSupport_{it}).$$

Potential empirical outcomes include aggregate measures of trauma-related symptoms, depression, anxiety, suicide mortality, health-service utilization, and long-run intergenerational consequences where ethically and historically appropriate data exist.

Such outcomes should be interpreted at the population level rather than as psychiatric diagnoses of political entities.

22 Philosophy and the Ethics of Demonstrated Power

The positive question is

What effects does demonstrated power produce?

The normative question is

When, if ever, is its use morally permissible?

These are not identical.

A consequentialist evaluation may examine

$$\Delta W = W(\text{intervention}) - W(\text{counterfactual}).$$

A deontological analysis instead asks whether actions satisfy independent moral constraints.

Just-war traditions introduce criteria including necessity, proportionality, discrimination, legitimate authority, and last resort.

The framework therefore explicitly rejects the inference

$$\text{causal effectiveness} \implies \text{moral legitimacy}.$$

Likewise,

$$\text{deterrence effectiveness} \not\Rightarrow \text{ethical permissibility}.$$

23 Physics: Distance, Reach, and Propagation

Physical distance matters because many effects propagate through space.

A generic inverse-power exposure model is

$$I(d) = \frac{K}{d^\rho}.$$

This motivates, but does not prove, a geopolitical distance-decay model.

Unlike idealized physical propagation, strategic effects depend on terrain, transport, detection, institutions, technology, and human decisions.

Effective strategic distance may therefore be represented by

$$d_{\text{eff}} = d \times \tau \times \ell \times \kappa,$$

where τ captures terrain, ℓ logistics, and κ defensive frictions.

The physical analogy motivates functional forms but cannot substitute for empirical estimation.

24 Chemistry: Thresholds and Strategic Activation

Chemical kinetics offers a useful mathematical analogy for threshold-dependent behavior.

Suppose the probability of political activation is

$$p_i = \frac{1}{1 + \exp[\gamma(E_i^* - R_i^S)]}.$$

Here E_i^* is a political activation threshold.

As strategic pressure exceeds the threshold,

$$R_i^S > E_i^*,$$

the probability of mobilization increases rapidly.

The analogy resembles an activation-energy problem, but political systems are not chemical systems. The value of the analogy lies in nonlinear modeling.

25 Biology and Evolutionary Adaptation

States and institutions can adapt to persistent environmental pressures.

Let institutional strategy x_i evolve according to

$$\dot{x}_i = \mu \frac{\partial U_i}{\partial x_i},$$

where $\mu > 0$ is an adaptation parameter.

A replicator-style model is

$$\dot{x}_k = x_k (\pi_k - \bar{\pi}),$$

where x_k is the prevalence of strategy k and π_k its payoff.

After a strategic shock, strategies suited to the new environment may expand.

Potential institutional adaptations include alliance formation, civil defense, industrial policy, military modernization, strategic stockpiling, technological investment, and diplomatic realignment.

26 Network Science

Geopolitical exposure need not be geographically local.

Let the international system be represented by graph

$$G = (V, E).$$

Let W be a weighted adjacency matrix.

A network exposure index is

$$N_i = \sum_j W_{ij} S_j.$$

Higher-order propagation gives

$$N = WS + \delta W^2 S + \delta^2 W^3 S + \dots.$$

For

$$|\delta| < \frac{1}{\rho(W)},$$

where $\rho(W)$ is the spectral radius,

$$N = (I - \delta W)^{-1} WS.$$

Thus countries geographically distant from SNoG may nevertheless be highly exposed through trade, finance, alliances, supply chains, or communications.

27 Information Theory

A demonstration is important partly because it contains information.

Let Θ denote uncertain SNoG capability and E the observed intervention.

The information revealed can be measured by mutual information:

$$I(\Theta; E) = \sum_{\theta, e} p(\theta, e) \log \frac{p(\theta, e)}{p(\theta)p(e)}.$$

A strategically surprising intervention has greater informational content than an event already predicted with probability close to one.

Bayesian surprise can be represented by the Kullback–Leibler divergence

$$D_{KL}(\pi_1 \| \pi_0) = \int \pi_1(\theta) \log \frac{\pi_1(\theta)}{\pi_0(\theta)} d\theta.$$

A possible empirical hypothesis is

$$|\Delta Policy_i| \propto D_{KL}(\pi_{1i} || \pi_{0i}).$$

28 Control Theory

A state can be represented abstractly as a dynamic system

$$x_{t+1} = Ax_t + Bu_t + Gw_t,$$

where

- x_t is the vector of economic and strategic state variables,
- u_t is policy,
- w_t is the external geopolitical shock.

The government chooses u_t to minimize

$$J = \sum_{t=0}^{\infty} \beta^t (x_t' Q x_t + u_t' R u_t).$$

A SNoG intervention alters either w_t or beliefs about its distribution. Policy adaptation is therefore a feedback-control problem.

29 Data Architecture

A credible empirical implementation requires several linked datasets.

Table 1: Proposed Empirical Variables

Category	Variable	Interpretation
Geography	Distance to SNoG	Geographic exposure to relevant SNoG nodes
Security	Defense expenditure/GDP	Fiscal military response
Security	Military personnel	Mobilization response
Finance	Sovereign spread	Market-implied geopolitical risk
Finance	Exchange-rate volatility	External financial instability
Investment	FDI inflows	Foreign investor response
Trade	Bilateral trade shares	Commercial realignment
Diplomacy	Voting similarity	Political alignment
Institutions	Executive constraints	Domestic political adaptation
Technology	Military R&D	Innovation response
Social	Migration/displacement	Population response
Health	Aggregate mental-health indicators	Human consequences of insecurity

Potential sources for historical validation include the Correlates of War project, SIPRI, World Bank, IMF, United Nations records, historical national accounts, archival military expenditure data, diplomatic archives, and financial-market databases where coverage permits.

30 Outcome Families

Rather than selecting outcomes after observing results, outcomes should be pre-specified by families. Define

$$Y_{it} = \begin{pmatrix} Y_{it}^{security} \\ Y_{it}^{fiscal} \\ Y_{it}^{financial} \\ Y_{it}^{trade} \\ Y_{it}^{diplomatic} \\ Y_{it}^{institutional} \\ Y_{it}^{social} \end{pmatrix}.$$

Multiple-testing corrections should accompany large outcome families.
For m hypotheses, a conservative Bonferroni threshold is

$$\alpha^* = \frac{\alpha}{m}.$$

False-discovery-rate procedures may offer greater power when many related hypotheses are examined.

31 Spatial Econometrics

Ordinary standard errors may be misleading when geographically adjacent states experience correlated shocks.

A spatial autoregressive model is

$$Y_t = \rho_W W Y_t + X_t \beta + \varepsilon_t.$$

Alternatively, spatial error dependence is

$$\varepsilon_t = \lambda_W W \varepsilon_t + u_t.$$

Inference can also employ distance-based covariance estimators.
The substantive point is simple:

$$\text{Cov}(\varepsilon_i, \varepsilon_j) \neq 0$$

may remain plausible when i and j are strategically connected.

32 Randomization Inference with Pseudo-SNoG Locations

A strong falsification exercise relocates the hypothetical treatment source.

For placebo iteration b , generate pseudo-location L_b .

Construct

$$P_i^{(b)} = g(d(i, L_b)).$$

Estimate

$$\hat{\beta}^{(b)}.$$

Compare the true estimate $\hat{\beta}^{SNoG}$ with the placebo distribution.

A randomization-inference p -value is

$$p_{RI} = \frac{1 + \sum_{b=1}^B \mathbf{1} \left\{ |\hat{\beta}^{(b)}| \geq |\hat{\beta}^{SNoG}| \right\}}{B + 1}.$$

If the actual geographic configuration produces an effect that is common under arbitrary pseudo-locations, the SNoG interpretation is weakened.

33 Placebo Dates

The same principle applies temporally.

Let T_b be placebo treatment dates.

Estimate

$$Y_{it} = \alpha_i + \lambda_t + \beta_b P_i \mathbf{1}\{t \geq T_b\} + \varepsilon_{it}.$$

A convincing design should not generate comparable effects for many arbitrary pre-treatment dates.

34 Synthetic Control Methods

For particularly exposed countries, synthetic controls can complement the panel design.

Let country 1 be highly exposed.

Construct weights

$$w_j \geq 0, \quad \sum_{j=2}^{J+1} w_j = 1.$$

Choose

$$w^* = \arg \min_w \sum_{t < T_0} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt} \right)^2.$$

The post-treatment gap is

$$\tau_t = Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt}.$$

This permits country-specific historical case studies without abandoning counterfactual discipline.

35 Machine Learning and Heterogeneous Treatment Effects

The treatment effect may vary across states.

Let

$$\tau(X_i) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i].$$

Relevant moderators include

$$X_i = (P_i, A_i, GDP_i, Regime_i, Trade_i, Nuclear_i, Alliance_i, Technology_i).$$

Causal forests or other flexible learners may estimate heterogeneous treatment effects, but machine learning should complement rather than replace the identification design.

A generic orthogonalized model is

$$Y_i - m(X_i) = \tau(X_i)[D_i - e(X_i)] + \varepsilon_i.$$

Here

$$m(X_i) = \mathbb{E}[Y_i \mid X_i],$$

and

$$e(X_i) = \mathbb{E}[D_i \mid X_i].$$

Cross-fitting can reduce overfitting bias.

36 Double Machine Learning

Consider

$$Y = \theta D + g(X) + U,$$

and

$$D = m(X) + V.$$

Estimate nuisance functions $\hat{g}(X)$ and $\hat{m}(X)$ using flexible methods.

Construct residuals

$$\tilde{Y} = Y - \hat{g}(X),$$

$$\tilde{D} = D - \hat{m}(X).$$

Then estimate

$$\tilde{Y} = \theta \tilde{D} + \epsilon.$$

The purpose is not to “let AI discover causality.” The causal interpretation continues to depend on research design and identifying assumptions.

37 A Bayesian SNoG Model

Let the latent strategic type of SNoG be

$$\theta \in \{\theta_L, \theta_H\}.$$

Prior belief is

$$\Pr(\theta = \theta_H) = p_0.$$

Suppose intervention probabilities satisfy

$$\Pr(E \mid \theta_H) = q_H,$$

$$\Pr(E \mid \theta_L) = q_L,$$

with

$$q_H > q_L.$$

Bayes’ rule gives

$$p_1 = \Pr(\theta_H \mid E) = \frac{q_H p_0}{q_H p_0 + q_L (1 - p_0)}.$$

Proposition 2 (Demonstration Raises Posterior Belief). *If*

$$0 < p_0 < 1$$

and

$$q_H > q_L > 0,$$

then

$$p_1 > p_0.$$

Proof. We require

$$\frac{q_H p_0}{q_H p_0 + q_L(1 - p_0)} > p_0.$$

Because the denominator is positive,

$$q_H p_0 > p_0 q_H p_0 + p_0 q_L(1 - p_0).$$

For $p_0 > 0$,

$$q_H > q_H p_0 + q_L(1 - p_0).$$

Therefore,

$$(1 - p_0)q_H > (1 - p_0)q_L.$$

Since $1 - p_0 > 0$, this is equivalent to

$$q_H > q_L,$$

which holds by assumption. □

This provides a formal basis for the demonstrated-power mechanism.

38 A General Equilibrium Extension

Country-level responses may alter global equilibrium.

Let country i choose

$$x_i = (C_i, I_i, D_i, A_i).$$

Its welfare depends on other states:

$$U_i = U_i(x_i, x_{-i}, P_i, \theta).$$

A Nash equilibrium satisfies

$$x_i^* \in \arg \max_{x_i} U_i(x_i, x_{-i}^*, P_i, \theta)$$

for every i .

A SNoG shock changes beliefs about θ , generating

$$x^*(\theta_0) \rightarrow x^*(\theta_1).$$

The aggregate effect is therefore not simply

$$\sum_i \text{partial-equilibrium effect}_i.$$

Trade diversion, alliance substitution, arms competition, migration, capital flows, and commodity-market responses can create general-equilibrium feedback.

39 Arms-Race Dynamics

Consider two states with military stocks M_1 and M_2 .

A Richardson-style system is

$$\dot{M}_1 = a_1 M_2 - b_1 M_1 + g_1 R_1,$$

$$\dot{M}_2 = a_2 M_1 - b_2 M_2 + g_2 R_2.$$

A SNoG demonstration can shift perceived threat R_i .

The system matrix is

$$A = \begin{pmatrix} -b_1 & a_1 \\ a_2 & -b_2 \end{pmatrix}.$$

Stability requires the real parts of the eigenvalues to be negative.

A sufficient condition is

$$b_1 b_2 > a_1 a_2.$$

Thus strategic shocks may either converge toward a new equilibrium or amplify arms competition depending on reaction parameters.

40 Technology and Innovation

Demonstrated military capability may alter research incentives.

Let research expenditure be R_i and technological stock T_i .

Suppose

$$T_{i,t+1} = (1 - \delta)T_{it} + \phi R_{it}^\alpha.$$

The government chooses research according to

$$R_{it}^* = R(P_i, \theta, A_i).$$

A demonstration may increase investment in

- military technology,
- nuclear technology,
- missile defense,
- civil defense,
- communications,
- intelligence,
- cybersecurity,
- energy security,
- strategic transportation infrastructure.

This produces long-run effects that can persist even if the original shock is temporary.

41 Supply Chains and Economic Geography

Let firm f choose production location l .

Expected profit is

$$\Pi_f(l) = \text{Revenue}_f(l) - \text{Cost}_f(l) - \text{Risk}_f(l).$$

Let geopolitical risk satisfy

$$\text{Risk}_f(l) = \psi P_l \text{Post}_t.$$

The optimal location is

$$l_f^* = \arg \max_l \Pi_f(l).$$

A strategic shock may therefore reallocate production, inventories, ports, warehouses, financial centers, and transportation networks.

The long-run geographic equilibrium may consequently differ from the short-run treatment effect.

42 Migration and Demography

Let migration utility from origin i to destination j be

$$U_{ij} = w_j - w_i - c_{ij} - \gamma(R_j - R_i).$$

Migration occurs if

$$U_{ij} > 0.$$

If proximity raises perceived insecurity,

$$\frac{\partial \text{Migration}_i}{\partial P_i} > 0$$

may follow.

Population movement can then alter labor markets, fiscal capacity, housing, political coalitions, and long-run growth.

43 An Integrated Structural System

The interdisciplinary mechanisms can be summarized by

$$B_{it} = \mathcal{B}(E_t, \mathcal{I}_{it}),$$

$$R_{it} = \mathcal{R}(P_i, B_{it}, A_i),$$

$$\text{Policy}_{it} = \mathcal{P}(R_{it}, X_{it}),$$

$$\text{Markets}_{it} = \mathcal{M}(\text{Policy}_{it}, R_{it}, \text{Global}_t),$$

$$\text{Welfare}_{it} = \mathcal{W}(\text{Policy}_{it}, \text{Markets}_{it}, \text{Security}_{it}, \text{Social}_{it}).$$

The causal chain is

$$\boxed{E_t \rightarrow B_{it} \rightarrow R_{it} \rightarrow \text{Policy}_{it} \rightarrow \text{Markets}_{it} \rightarrow \text{Welfare}_{it}.}$$

44 Directed Acyclic Graph

Figure 1 summarizes the identification logic.

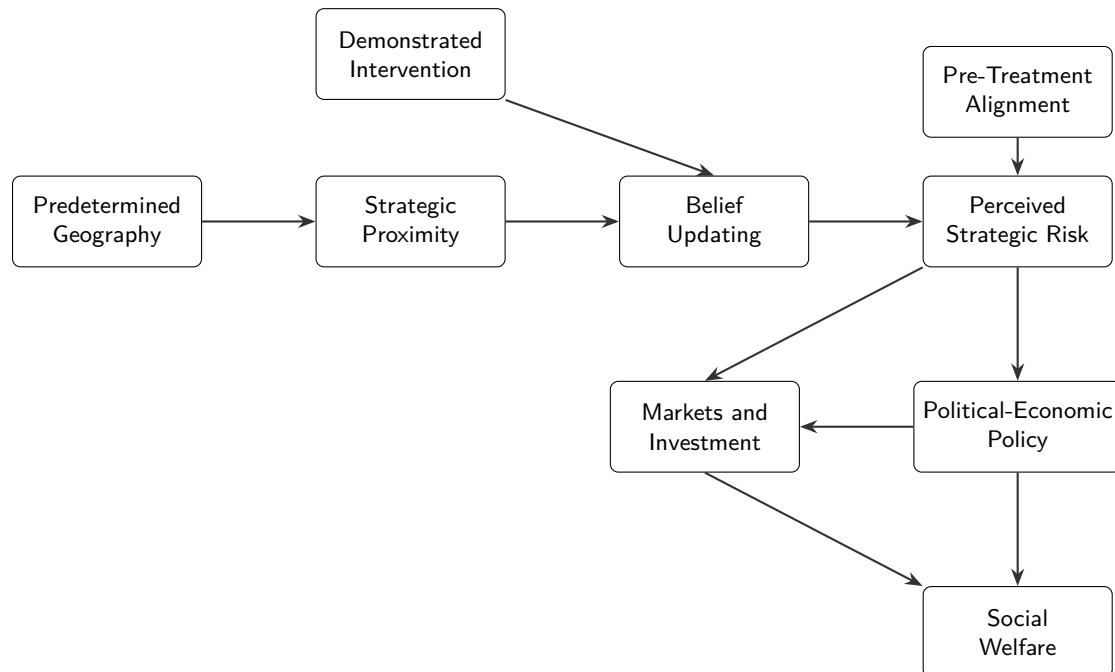


Figure 1: Conceptual causal architecture of demonstrated power and strategic proximity.

45 Vector Graphic: Distance-Decay Function

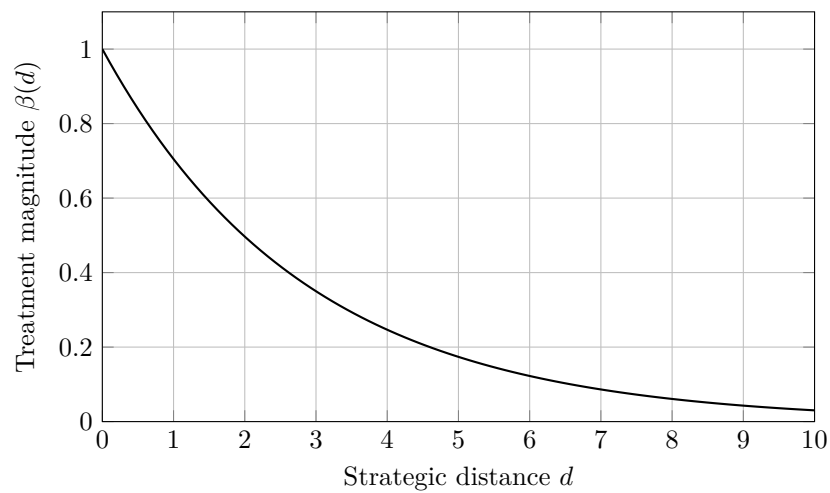


Figure 2: Illustrative exponential distance decay, $\beta(d) = \beta_0 e^{-\rho d}$, normalized to $\beta_0 = 1$.

46 Vector Graphic: Threat versus Protection

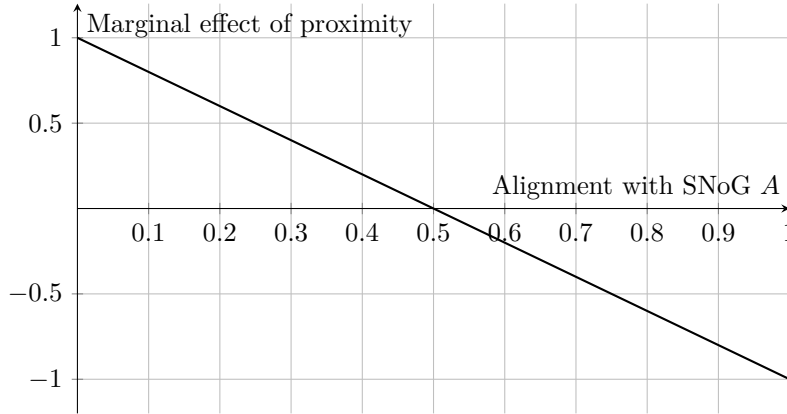


Figure 3: Illustrative transition from threat-dominated to protection-dominated proximity effects.

47 Empirical Hypotheses

The framework produces the following testable hypotheses.

Table 2: Principal Empirical Hypotheses

Hypothesis	Prediction
H1	Greater adversarial SNoG proximity increases defense expenditure after a salient demonstration.
H2	Greater proximity changes sovereign risk premia, exchange-rate volatility, capital flows, and investment.
H3	Exposed governments alter diplomatic and alliance behavior.
H4	Additional military expenditure produces measurable fiscal substitution.
H5	Political and administrative institutions adapt to the changed security environment.
H6	Treatment magnitude decays with strategic rather than merely Euclidean distance.
H7	Credible SNoG protection can reverse the sign of the defense response.
H8	More surprising demonstrations generate larger responses through belief updating.
H9	Effects propagate through trade, alliance, financial, and communications networks.
H10	Persistent strategic exposure generates technological and institutional adaptation.

48 Historical Validation Strategy

The Hiroshima–Nagasaki episode can motivate a historical validation exercise, but the empirical design must be constructed carefully.

A naive specification such as

$$Y_{it} = \alpha_i + \lambda_t + \beta \text{DistanceUSA}_i \times \text{Post1945}_t + \varepsilon_{it}$$

faces major identification problems because 1945 coincides with the end of World War II, enormous destruction, occupation, border changes, decolonization, institutional reconstruction, and subsequent Cold War formation.

Accordingly, the historical exercise should not mechanically attribute all post-1945 spatial differences to nuclear demonstration.

A more credible design should incorporate

- wartime exposure,
- Axis/Allied/neutral status,
- occupation status,
- pre-war economic trends,
- destruction intensity,
- regional fixed effects,
- country-specific trends where appropriate,
- alliance status,
- subsequent Soviet nuclear acquisition,
- Korean War exposure,
- decolonization,
- Marshall Plan and reconstruction effects.

The historical analysis is therefore best viewed as a demanding validation exercise rather than an automatic natural experiment.

49 Threats to Identification

The principal threats are summarized below.

Table 3: Identification Threats and Responses

Threat	Problem	Proposed Response
Differential trends	Near and far states already evolve differently	Event-study pre-trends
Anticipation	Governments react before T_0	Leads and alternative event windows
Spatial correlation	Neighboring errors are correlated	Spatially robust inference
Concurrent shocks	Other events coincide with T_0	Region-by-time effects and controls
Treatment endogeneity	Strategic nodes are deliberately located	Predetermined geography and historical instruments
Spillovers	Control states are indirectly treated	Network exposure models
Outcome selection	Researcher selects favorable variables	Pre-specified outcome families
Functional-form dependence	Results depend on $1/d$	Alternative distance kernels
Outliers	A few countries drive estimates	Leave-one-out analysis
False geography	Any arbitrary location generates effects	Pseudo-location randomization inference

50 Robustness Architecture

A credible paper should report at least the following classes of robustness checks:

1. alternative proximity measures;
2. alternative strategic nodes;
3. alternative treatment dates;
4. event-study leads and lags;
5. region-by-year fixed effects;
6. country-specific trends;
7. spatially robust inference;
8. clustered standard errors;
9. randomization inference;
10. pseudo-SNoG locations;
11. leave-one-country-out estimation;
12. matching or entropy balancing;
13. synthetic controls;
14. nonlinear distance decay;
15. alternative alignment measures;
16. heterogeneous treatment effects;
17. multiple-testing corrections;
18. sensitivity to concurrent geopolitical events.

51 Pseudo-Code for the Empirical Pipeline

Algorithm 1: SNoG Proximity Quasi-Experiment

1. Define the intervention event T_0 .
2. Define SNoG strategic nodes $j = 1, \dots, J$.
3. For every country i , calculate d_{ij} .
4. Construct
$$P_i = \sum_j \frac{w_j}{(d_{ij} + d_0)^\rho}.$$
5. Freeze all treatment definitions before examining outcomes.
6. Assemble pre-treatment and post-treatment panel data.
7. Plot raw trends by proximity quantile.
8. Estimate the baseline fixed-effects model.
9. Estimate the event-study model.
10. Test pre-treatment coefficients.
11. Estimate alignment interactions.

12. Estimate nonlinear distance-decay functions.
13. Re-estimate using alternative geographic definitions.
14. Apply spatially robust inference.
15. Generate placebo treatment dates.
16. Generate pseudo-SNoG geographic locations.
17. Construct randomization-inference distributions.
18. Conduct leave-one-country-out estimation.
19. Estimate synthetic controls for highly exposed cases.
20. Estimate heterogeneous treatment effects.
21. Correct for multiple hypothesis testing.
22. Separate statistical significance from economic magnitude.
23. Conduct welfare analysis.
24. Report null results as well as statistically significant results.

52 Pseudo-Code for AI-Assisted Heterogeneity Analysis

Algorithm 2: Causal Machine-Learning Extension

1. Partition observations into K folds.
2. For each fold k :
 - (a) estimate the outcome nuisance function $m(X)$ using the other folds;
 - (b) estimate the treatment nuisance function $e(X)$;
 - (c) predict both functions on fold k ;
 - (d) construct orthogonalized outcome and treatment residuals.
3. Pool cross-fitted residuals.
4. Estimate the average orthogonalized treatment effect.
5. Estimate $\tau(X)$ using a heterogeneous-treatment learner.
6. Partition estimated effects by:
 - (a) alignment,
 - (b) income,
 - (c) regime type,
 - (d) trade dependence,
 - (e) military capability,
 - (f) nuclear status,
 - (g) geographic proximity,
 - (h) network proximity.
7. Test whether discovered heterogeneity replicates out of sample.
8. Treat machine-learning findings as exploratory unless pre-specified or independently validated.

53 Simulation Framework

Before empirical estimation, simulation can determine whether the proposed design has adequate power.

Generate

$$Y_{it} = \alpha_i + \lambda_t + \beta P_i Post_t + \rho Y_{i,t-1} + u_{it}.$$

Let spatial errors satisfy

$$u_t = \lambda W u_t + \epsilon_t.$$

Monte Carlo experiments can vary

$$\beta, \rho, \lambda, N, T$$

and the distribution of P_i .

For replication b , estimate $\hat{\beta}_b$.

Bias is

$$Bias(\hat{\beta}) = \frac{1}{B} \sum_{b=1}^B (\hat{\beta}_b - \beta).$$

Root mean squared error is

$$RMSE = \sqrt{\frac{1}{B} \sum_{b=1}^B (\hat{\beta}_b - \beta)^2}.$$

Statistical power is

$$Power = \Pr(\text{reject } H_0 : \beta = 0 \mid \beta \neq 0).$$

54 Falsifiability

The SNoG proximity hypothesis must be falsifiable.

Evidence against the theory would include:

1. strong differential pre-trends;
2. no post-intervention proximity gradient;
3. effects appearing equally under arbitrary placebo dates;
4. effects appearing under arbitrary pseudo-SNoG locations;
5. instability under reasonable proximity measures;
6. absence of predicted alignment heterogeneity;
7. outcomes moving before beliefs could plausibly update;
8. stronger explanatory power from unrelated concurrent shocks.

A theoretical framework that cannot fail an empirical test cannot provide a strong causal explanation.

55 Interpretation of Null Results

A null estimate

$$\hat{\beta} \approx 0$$

does not necessarily imply that demonstrated power is irrelevant.

Possible explanations include

- the intervention was already anticipated;
- proximity is measured poorly;
- alliances neutralize geographic exposure;
- effects operate through networks rather than physical distance;
- responses occur at horizons outside the sample;
- heterogeneous positive and negative effects cancel;
- institutions prevent rapid policy adjustment;
- statistical power is inadequate.

These alternatives should be distinguished empirically where possible.

56 External Validity

The general framework is not restricted to SNoG.

It applies whenever

predetermined exposure $\times$ unexpected institutional or strategic shock

generates plausibly exogenous variation.

Possible applications include

- military alliances,
- nuclear umbrellas,
- foreign military bases,
- missile-defense systems,
- sanctions regimes,
- trade blocs,
- maritime-security arrangements,
- strategic infrastructure,
- technological demonstrations,
- major discoveries affecting military capabilities.

The broader proposition is therefore a theory of the political economy of strategic proximity.

57 Limits of the Hiroshima–Nagasaki Analogy

Historical analogies are useful only when their limits are explicit.

First, the bombings occurred during an ongoing world war.

Second, the United States briefly possessed a nuclear monopoly, but the international nuclear environment changed rapidly.

Third, the post-war international system simultaneously experienced occupation, reconstruction, decolonization, institutional formation, ideological competition, and technological change.

Fourth, nuclear deterrence subsequently depended on second-strike capability, command-and-control systems, delivery technologies, alliance commitments, and mutual vulnerability.

Fifth, contemporary SNoG applications may involve fundamentally different institutions, technologies, legal constraints, and political environments.

Consequently,

historical analogy $\neq$ causal identification.

The analogy motivates hypotheses. Data and research design must test them.

58 Ethical and Legal Boundaries

A political-economic analysis of nuclear warfare must distinguish description, prediction, and prescription.

This paper studies

effects of demonstrated strategic capability,

not operational methods for causing destruction.

International humanitarian law, civilian protection, proportionality, distinction, and the catastrophic humanitarian consequences of nuclear weapons are integral to normative evaluation.

The scientific value of studying deterrence and geopolitical risk does not require endorsement of warfare.

59 A Unified SNoG Proximity Index

A comprehensive index can combine physical and network exposure.

Define

$$Z_i = \begin{pmatrix} P_i^{geo} \\ P_i^{trade} \\ P_i^{finance} \\ P_i^{alliance} \\ P_i^{logistics} \\ P_i^{information} \end{pmatrix}.$$

Then

$$SPI_i = \omega' Z_i,$$

subject to

$$\omega_k \geq 0, \quad \sum_k \omega_k = 1.$$

Alternatively, weights may be estimated from pre-treatment data or derived through principal-component methods.

The index should not be optimized to maximize the estimated treatment effect, because doing so would create severe specification-search bias.

60 A Structural SNoG Response Function

Let

$$\mathcal{S}_i = P_i^\alpha C_i^\beta T_i^\gamma A_i^\delta N_i^\eta,$$

where

- P_i is strategic proximity,
- C_i demonstrated capability,
- T_i perceived threat,
- A_i alignment,
- N_i network exposure.

Taking logarithms,

$$\log \mathcal{S}_i = \alpha \log P_i + \beta \log C_i + \gamma \log T_i + \delta \log A_i + \eta \log N_i.$$

This representation permits elasticity interpretation.

For example,

$$\frac{\partial \log \mathcal{S}_i}{\partial \log P_i} = \alpha.$$

61 Main Theoretical Result

Theorem 1 (Demonstrated-Power Proximity Theorem). *Suppose:*

1. *geographic proximity P_i is predetermined;*
2. *an intervention E raises the posterior expected capability or willingness parameter:*

$$\mathbb{E}[\theta \mid E] > \mathbb{E}[\theta];$$

3. *perceived strategic exposure satisfies*

$$R_i = P_i \mathbb{E}[\theta \mid E] \Xi_i;$$

4. *optimal policy a_i^* is differentiable in R_i .*

Then the post-demonstration marginal policy response to proximity is

$$\frac{\partial a_i^*}{\partial P_i} = \frac{\partial a_i^*}{\partial R_i} \mathbb{E}[\theta \mid E] \Xi_i.$$

Therefore, the sign of the proximity effect depends jointly on the policy response to strategic exposure and the sign of the state's strategic relationship with SNoG.

Proof. By assumption,

$$R_i = P_i \mathbb{E}[\theta \mid E] \Xi_i.$$

Holding posterior beliefs and alignment fixed,

$$\frac{\partial R_i}{\partial P_i} = \mathbb{E}[\theta \mid E] \Xi_i.$$

Applying the chain rule,

$$\frac{\partial a_i^*}{\partial P_i} = \frac{\partial a_i^*}{\partial R_i} \frac{\partial R_i}{\partial P_i}.$$

Substitution gives

$$\frac{\partial a_i^*}{\partial P_i} = \frac{\partial a_i^*}{\partial R_i} \mathbb{E}[\theta \mid E] \Xi_i.$$

Hence geographic proximity alone does not determine the sign. The effect is mediated by posterior beliefs and the strategic relationship.

□

Corollary 1 (Threat Regime). *If*

$$\Xi_i > 0$$

and

$$\frac{\partial D_i^*}{\partial R_i} > 0,$$

then

$$\frac{\partial D_i^*}{\partial P_i} > 0.$$

Corollary 2 (Protection Regime). *If the relevant exposure variable represents a credible security guarantee and domestic defense is a substitute for external protection, then greater proximity may satisfy*

$$\frac{\partial D_i^*}{\partial P_i} < 0.$$

62 Interpretive Matrix

Table 4: Possible Political-Economic Regimes

Proximity	Relationship	Predicted Mechanism
High	Adversarial	Threat and defensive mobilization
High	Allied	Security umbrella and possible defense substitution
High	Neutral	Hedging and strategic ambiguity
Low	Adversarial	Lower immediate exposure but possible balancing
Low	Allied	Weaker direct security externality
Low	Neutral	Limited direct response

63 Research Design Decision Tree

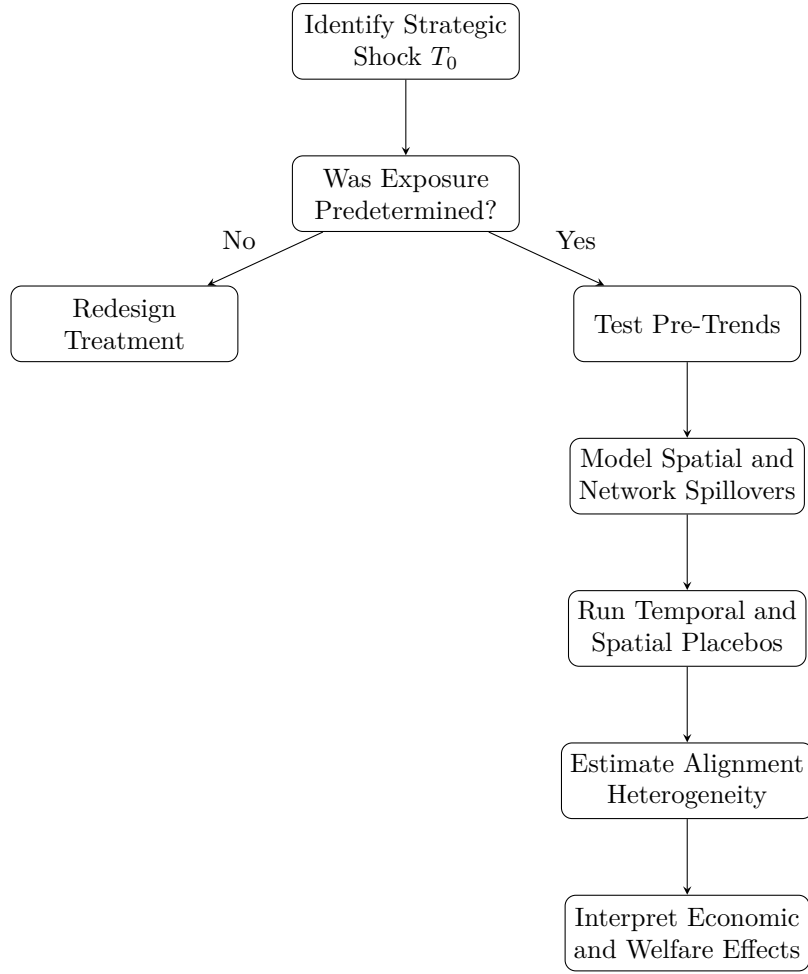


Figure 4: Research-design decision tree for a SNoG proximity study.

64 Discussion

The principal insight of the framework is that strategic geography is state-dependent.

Physical distance can remain constant while its political-economic meaning changes because governments update beliefs about military capability, willingness to intervene, alliance credibility, and technological possibility.

The Hiroshima and Nagasaki bombings provide an historically important example of demonstrated capability. After August 1945, nuclear weapons could no longer be treated solely as speculative technology. Yet the subsequent international order also demonstrates why simple causal stories are dangerous. Nuclear proliferation, Soviet-American competition, alliance systems, economic reconstruction, decolonization, and institutional formation rapidly transformed the strategic environment.

The SNoG framework therefore emphasizes conditional effects rather than universal claims.

The key equation is not

$$P_i \uparrow \implies Y_i \uparrow .$$

It is

$$\frac{\partial Y_i}{\partial P_i} = f(\text{demonstrated capability, beliefs, alignment, institutions, networks, economic structure}).$$

This formulation is simultaneously more difficult to estimate and more realistic.

65 Conclusion

This paper develops a theory and empirical architecture for the political economy of SNoG proximity. Its central proposition is

$$\boxed{\text{Geography} \times \text{Demonstrated Power} \times \text{Strategic Relationship} \rightarrow \text{Political-Economic Adjustment.}}$$

The Hiroshima–Nagasaki episode motivates the distinction between latent and demonstrated capability. A salient intervention may alter beliefs even when geography and underlying material resources remain unchanged.

This produces a quasi-experimental strategy in which predetermined strategic proximity is interacted with a discrete information-generating intervention.

The framework generates several empirical predictions concerning military expenditure, fiscal composition, sovereign risk, investment, trade, diplomatic alignment, technological innovation, migration, institutional adaptation, and social welfare.

The most important theoretical qualification is that proximity has no intrinsically positive or negative sign.

For an adversarial state,

$$\text{proximity} \rightarrow \text{threat.}$$

For an allied state,

$$\text{proximity} \rightarrow \text{protection.}$$

For a neutral state,

$$\text{proximity} \rightarrow \text{hedging.}$$

Consequently, the political economy of SNoG proximity is fundamentally a political economy of demonstrated power, strategic distance, beliefs, networks, and endogenous adaptation.

The empirical standard must remain demanding. Historical analogy is not causal evidence. Geographic persistence is not sufficient for identification. Statistical significance is not equivalent to strategic importance, and strategic effectiveness is not equivalent to moral legitimacy.

A successful empirical implementation must therefore combine theory, historical institutional knowledge, transparent treatment construction, event-study diagnostics, spatial inference, falsification tests, randomization inference, heterogeneous treatment analysis, and welfare evaluation.

The resulting research program extends beyond SNoG. It provides a general framework for studying how the demonstration of concentrated strategic power changes the economic and political value of location itself.

References

- [1] Acemoglu, D., Johnson, S., and Robinson, J. A. (2005). Institutions as a fundamental cause of long-run growth. In P. Aghion and S. N. Durlauf (eds.), *Handbook of Economic Growth*, Vol. 1A. Elsevier.
- [2] Angrist, J. D., and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- [3] Aronow, P. M., Samii, C., and Assenova, V. A. (2017). Cluster-robust variance estimation for dyadic data. *Political Analysis*, 23, 564–577.
- [4] Ashworth, S., Berry, C. R., and Bueno de Mesquita, E. (2017). Theory and credibility: Integrating theoretical and empirical social science. *Journal of Politics*, 79, 1189–1204.
- [5] Athey, S., and Imbens, G. W. (2019). Machine learning methods that economists should know about. *Annual Review of Economics*, 11, 685–725.
- [6] Barro, R. J. (1990). Government spending in a simple model of endogenous growth. *Journal of Political Economy*, 98, S103–S125.

- [7] Belloni, A., Chernozhukov, V., and Hansen, C. (2014). Inference on treatment effects after selection among high-dimensional controls. *Review of Economic Studies*, 81, 608–650.
- [8] Bertrand, M., Duflo, E., and Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics*, 119, 249–275.
- [9] Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *Econometrica*, 86, C1–C68.
- [10] Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92, 1–45.
- [11] Fearon, J. D. (1995). Rationalist explanations for war. *International Organization*, 49, 379–414.
- [12] Fisher, R. A. (1935). *The Design of Experiments*. Oliver and Boyd.
- [13] Hirshleifer, J. (1995). Anarchy and its breakdown. *Journal of Political Economy*, 103, 26–52.
- [14] Imbens, G. W., and Rubin, D. B. (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences*. Cambridge University Press.
- [15] Jervis, R. (1978). Cooperation under the security dilemma. *World Politics*, 30, 167–214.
- [16] Kahneman, D., and Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47, 263–291.
- [17] Myerson, R. B. (1991). *Game Theory: Analysis of Conflict*. Harvard University Press.
- [18] Nash, J. F. (1950). Equilibrium points in n -person games. *Proceedings of the National Academy of Sciences*, 36, 48–49.
- [19] Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*. Second edition. Cambridge University Press.
- [20] Richardson, L. F. (1960). *Arms and Insecurity: A Mathematical Study of the Causes and Origins of War*. Boxwood Press.
- [21] Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66, 688–701.
- [22] Schelling, T. C. (1960). *The Strategy of Conflict*. Harvard University Press.
- [23] Schelling, T. C. (1966). *Arms and Influence*. Yale University Press.
- [24] Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27, 379–423 and 623–656.
- [25] Spirling, A. (2012). US treaty-making with American Indians: Institutional change and relative power, 1784–1911. *American Journal of Political Science*, 56, 84–97.
- [26] Tversky, A., and Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5, 297–323.
- [27] Waltz, K. N. (1979). *Theory of International Politics*. Addison-Wesley.
- [28] Wiener, N. (1948). *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press.
- [29] World Bank. *World Development Indicators*. World Bank, Washington, DC.
- [30] Stockholm International Peace Research Institute. *SIPRI Military Expenditure Database*. SIPRI, Stockholm.

Glossary

SNoG

Standard Nuclear oliGARCHy, the strategic institutional system whose geographic and political-economic proximity is studied in this paper.

SNoG Proximity

A measure of geographic, logistical, economic, financial, alliance, or network exposure to SNoG.

Demonstrated Power

Capability whose operational existence or willingness of use becomes more credible through observable action.

Latent Capability

Strategic capacity whose existence, effectiveness, or probability of use remains incompletely observed.

Strategic Distance

A generalized distance measure incorporating geography, logistics, economics, networks, and political relationships.

Strategic Proximity Index

A composite measure summarizing multiple dimensions of exposure to SNoG.

Quasi-Natural Experiment

An observational setting in which institutional or historical circumstances generate plausibly exogenous treatment variation.

Difference-in-Differences

A causal-inference design comparing changes across units with different treatment exposure.

Continuous Treatment

A treatment whose intensity varies continuously rather than taking only treated and untreated values.

Event Study

A dynamic regression examining treatment effects before and after an event.

Parallel Trends

The identifying condition that differently exposed units would have followed comparable counterfactual trends absent treatment.

Strategic Alignment

The extent to which a state is politically, militarily, or institutionally aligned with SNoG.

Security Externality

Security benefits or costs imposed on one state by another state's military capabilities or commitments.

Deterrence

Prevention of an action by altering the expected costs and benefits associated with that action.

Bayesian Updating

Revision of probability beliefs after observing new information.

Bayesian Surprise

The divergence between posterior and prior beliefs following an observation.

Spatial Spillover

An effect transmitted from one geographic unit to another.

Network Spillover

An effect transmitted through trade, financial, alliance, information, or other networks.

Randomization Inference

Inference obtained by comparing an observed statistic with a distribution generated under placebo or randomized treatment assignments.

Pseudo-SNoG Location

A placebo strategic location used to test whether the observed geographic effect is unusual relative to arbitrary locations.

Synthetic Control

A weighted combination of comparison units constructed to approximate the pre-treatment trajectory of an exposed unit.

Heterogeneous Treatment Effect

A causal effect that differs according to observed characteristics.

Double Machine Learning

A causal estimation framework using flexible nuisance-function estimation and orthogonalized moment conditions.

Warfare Economics

The study of resource allocation, production, finance, mobilization, and economic behavior under military conflict or preparation for conflict.

General Equilibrium

An equilibrium in which responses across multiple markets and actors are determined jointly.

Strategic Hedging

Policies intended to preserve options when a state faces uncertainty regarding future threats, alliances, or distributions of power.

Distance Decay

The reduction in treatment intensity or strategic influence as effective distance increases.

Half-Life Distance

The distance at which an estimated proximity effect falls to one-half of its value at the origin.

Mutual Information

An information-theoretic measure of statistical dependence between random variables.

Kullback–Leibler Divergence

A measure of the informational difference between two probability distributions.

Control Theory

The mathematical study of dynamic systems whose states are influenced through feedback and policy controls.

Replicator Dynamics

An evolutionary model in which strategies with above-average payoffs increase their population shares.

Potential Outcome

The outcome that would occur for a unit under a particular treatment state.

Falsification Test

An empirical test designed to identify patterns that should not occur if the proposed causal explanation is correct.

The End